



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

UNIT TEST

21-11-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : TEPS, TOPS, I(A), I, I(B), II(A), II & LIVE-I, I(B), II(A)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	C	C	C	C	A	C	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	B	A	A	B	A	C	B	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	2.00	5.00	160.00	2.00	8.00	8.00	6.00	0.04	1751.00 to 1752.00	3.33

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	B	C	B	A	D	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	A	D	A	D	D	C	C	D	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	4.00	9.00	3.00	3.00	3.00	180.00	1.34	3.00	1.55 to 1.56	3.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	A	D	A	D	D	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	C	C	C	C	C	B	D	D	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1.00	2.00	5.00	2.00	4.00	6.00	2.00	4.00	64.00	7.00

HINT – SHEET

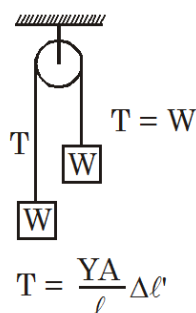
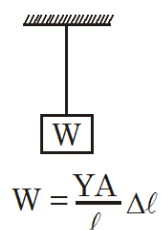
PART-1 : PHYSICS

SECTION-I

1. Ans (D)

$$\Delta l \propto \frac{l}{d^2}$$

2. Ans (C)



3. Ans (C)

The volume of the material remains the same as the fixed volume is drawn into a wire.

Fixed volume = V

The volume of the wire is given as :

$$V = A \times L$$

Here, A is the area of cross - section of wire and L is the length of wire.

$$\Delta l = x \rightarrow \text{extension}$$

$$\text{from the stress - strain relation : } \frac{F}{A} = Y \frac{\Delta l}{L}$$

$$\Delta l = \frac{F}{Y} \times \frac{L}{V} \times L; X = \frac{FL^2}{YV}$$

$$X \propto L^2$$

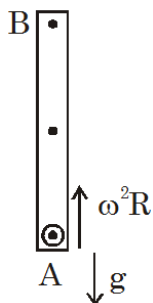
4. Ans (C)

$$KE_P = \frac{1}{2}m(2v)^2$$

$$KE_S = \frac{3}{4}mv^2$$

5. Ans (C)

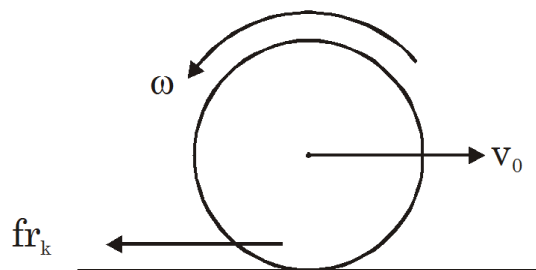
$$\theta = \omega t = \frac{\pi}{2} \text{ rad}$$



$$a_A = \omega^2 R \hat{j} - g \hat{j}$$

$$= \left(\frac{\pi}{2}\right)^2 \cdot \frac{L}{2} \hat{j} - 10 \hat{j} = \frac{\pi^2}{4} \cdot \frac{4}{2} \hat{j} - 10 \hat{j} = -5 \hat{j}$$

6. Ans (C)



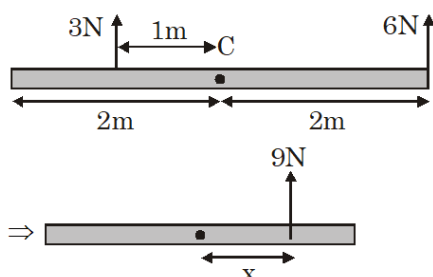
Let at time t total KE be comes zero.

$$\text{So, } 0 = v_0 - \frac{f_{rk}}{m}t \Rightarrow 0 = \omega_0 - \frac{f_{rk} \cdot R}{\left(\frac{mR^2}{2}\right)}t$$

$$\text{Hence } \omega_0 = \frac{2v_0}{R} = \frac{2 \times 10}{0.2}$$

$$\omega_0 = 100 \text{ rad/sec}$$

7. Ans (A)



$$F_{\text{net}} = 3 + 6 = 9 \text{ N}$$

$$(\tau_{\text{net}})_{\text{about c}} = 6 \times 2 - 3 \times 1 = 9$$

$$\tau_c = 9 \times x = 9$$

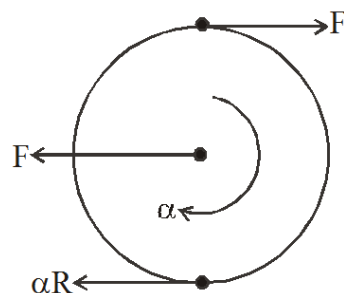
$$\Rightarrow x = 1 \text{ m}$$

8. Ans (C)

$$\alpha = \frac{F x}{\frac{mr^2}{4} + mx^2}$$

$$\text{For maximum } \alpha, \frac{d\alpha}{dx} = 0$$

9. Ans (A)



Net force

$$F - F = 0 = ma; a = 0$$

$$\therefore \tau = RF \text{ (clockwise)}$$

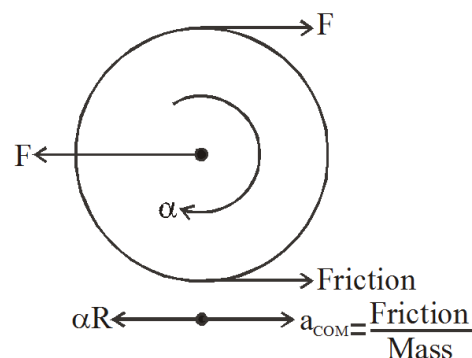
$\therefore \alpha$ would be clockwise and acceleration αR at the bottom most point would leftward.

$$[\because \vec{\tau} = I\vec{\alpha}]$$

For pure rolling, acceleration of bottom most point should be zero because surface is stationary.

So acceleration of C.O.M. should be rightward.

Hence friction should be rightward.



10. Ans (C)

$$\text{Impulse} = F \cdot \Delta t = \Delta P = \vec{P}_f - \vec{P}_i$$

$$= m(\vec{v}_f - \vec{v}_i)$$

Impulse of magnitude $2m$ in the negative x -directions is $-2m\hat{i}$

$$\therefore -2m\hat{i} = m[\vec{v}_f - (\hat{i} - 3\hat{j})]$$

$$-\vec{v}_f = -2\hat{i} + \hat{i} - 3\hat{j} = -\hat{i} - 3\hat{j}$$

11. **Ans (B)**

$$\text{Initial } V_{\text{cm}} = \frac{4 \times 1 + 2 \times 0}{1 + 2} = \frac{4}{3} \text{ m/s upwards}$$

$$a_{\text{cm}} = g \text{ downward}$$

$\therefore$ centre of mass first moves up and then comes down.

13. **Ans (B)**

At any instant of time let velocity of pail of chain which is just coming in contact with surface = v and

dx = length of chain reaches on surface in time dt then momentum carried by this part

$$dp = (dm) v$$

$\therefore$ rate of change of momentum of chain

$$\frac{dp}{dt} = \frac{dm}{dt} \cdot v = \lambda \frac{dx}{dt} v = \lambda v \cdot v = \left(\frac{m}{\ell} \right) v^2$$

$$\text{as } v = \sqrt{2g \left(\frac{\ell}{2} \right)} = \sqrt{g\ell} \text{ so}$$

$$\frac{dp}{dt} = \frac{m}{\ell} (g\ell) = mg$$

$$\text{weight of parts on the surface} = \frac{mg}{2}$$

$\therefore$ net force by chain on surface

$$= mg + \frac{mg}{2} = \frac{3}{2} mg$$

14. **Ans (A)**

$$\frac{mgh}{2} = ms\Delta T$$

$$\Rightarrow \Delta T = \frac{gH}{2s}$$

$$\Delta T = \frac{9.8 \times 84}{2 \times 4200}$$

$$\Delta T = 0.098^\circ\text{C}$$

17. **Ans (A)**

$$\text{Heat lost} = \text{Heat gained } 1 \times 4200 \times 80$$

$$= m_{\text{ice}} \times 336 \times 1000$$

$$\Rightarrow m_{\text{ice}} = 1 \text{ kg}$$

$\therefore$ Entire ice will melt

18. **Ans (C)**

$$T_B = 4T_0$$

$$T_D = 5T_0$$

$$T_E = 4T_0$$

$$T_F = 3T_0$$

19. **Ans (B)**

When temperature of rod is increased than

$$l_2 = l_1 [1 + \alpha(\theta)]$$

If ($l_3 \neq l_2$) than rod will compress or expand so stress will produced.

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.00)**

$$1 \times S_A (30 - 25) = 2 \times S_B \times (25 - 20)$$

$$\frac{S_A}{S_B} = 2$$

2. **Ans (5.00)**

$$\Delta Q_A \ell + \Delta Q_{\text{water}} = \Delta Q_{\text{iron}}$$

$$0.5 \times 1140 \times (T - 20) + 0.2 \times 4200 (T - 20)$$

$$= 0.2 \times 470 \times (100 - T)$$

$$T = 25^\circ\text{C}$$

3. **Ans (160.00)**

$$F = yA \frac{\Delta L}{L} = yA (\alpha \Delta T)$$

4. **Ans (2.00)**

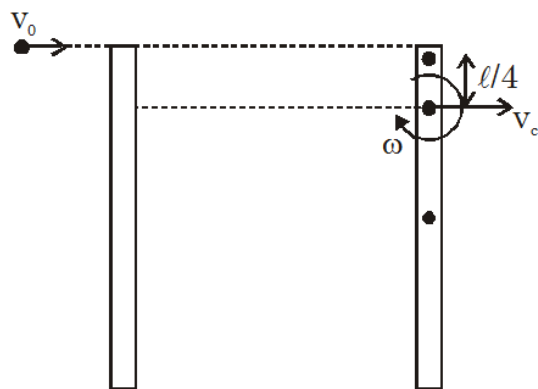
$$\Delta V = V_0 \times \gamma_{\text{app}} \times \Delta T$$

$$= V_0 (\gamma - 3\alpha) \times \Delta T$$

$$= 10^3 (7 - 3) \times 10^{-5} \times 50$$

$$= 2 \text{ ml}$$

5. Ans (8.00)

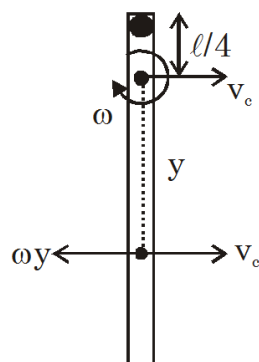


$$mv_0 = 2mv_c \dots (i)$$

$$mv_0 \frac{l}{4} = I_c \omega \dots (ii)$$

$$mv_0 \frac{l}{4} = \left[\frac{m l^2}{12} + m \left(\frac{l}{4} \right)^2 + m \left(\frac{l}{4} \right)^2 \right] \omega$$

$$\omega = \frac{6}{5} \frac{v_0}{l}$$

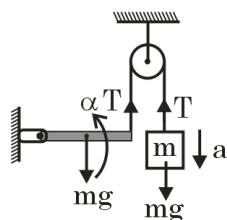


$$y = \frac{v_c}{\omega} = \frac{v_0/2}{\frac{6}{5} \left(\frac{v_0}{l} \right)}; y = \frac{5}{12} l$$

$$x_0 = \frac{l}{4} + y; x_0 = 80 \text{ cm};$$

$$\frac{x_0}{10} = 8 \text{ cm}$$

6. Ans (8.00)



$$mg - T = ma \dots (1)$$

$$TL - mg \frac{L}{2} = \frac{mL^2}{3} \alpha \dots (2)$$

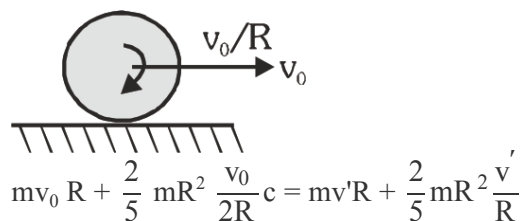
$$L\alpha = a \dots (3)$$

$$\text{Solving } a = \frac{3g}{8}$$

$$\Rightarrow k = 8$$

7. Ans (6.00)

By applying conservation of angular momentum



$$mv_0 R + \frac{2}{5} mR^2 \frac{v_0}{2R} = mv'R + \frac{2}{5} mR^2 \frac{v'}{R}$$

$$\Rightarrow \frac{6}{5} mv_0 R = \frac{7}{5} mv'R \Rightarrow v' = \frac{6}{7} v_0$$

8. Ans (0.04)

$$\omega_i = 0$$

$$\omega_f = 1800 \times \frac{2\pi}{60} \text{ rad/s} = 60\pi$$

$$\omega_f = \omega_i + \alpha t$$

$$\Rightarrow \alpha = 15\pi \text{ rad/s}^2$$

$$\tau = I\alpha = \frac{MR^2}{2} \alpha$$

$$\tau = 0.0471$$

9. Ans (1751.00 to 1752.00)

$$0.1 \text{ cm} = 9.9 \alpha_{\text{lead}} \Delta T - 10 \alpha_{\text{Al}} \Delta T$$

10. Ans (3.33)

Given point is point of percussion.

$$d = \frac{L}{2} - \frac{I_{\text{cm}}}{MR}$$

$$= \frac{L}{2} - \frac{ML^2/12}{(ML/2)} = \frac{L}{2} - \frac{L}{6}$$

$$d = \frac{L}{3}$$

$$d = \frac{10}{3} = 3.33 \text{ m}$$

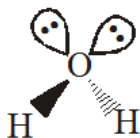
PART-2 : CHEMISTRY

SECTION-I

10. **Ans (D)**

A planar sp^3 like H_2O

one s-orbital, three P-orbital



100% s 300% p

1 bond 18%

2 bond 36%

Remaining will be distribute between lone pair

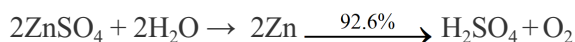
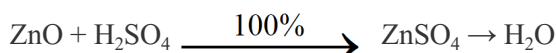
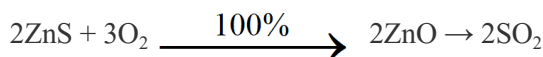
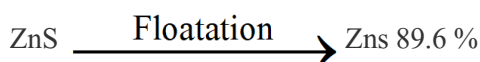
$$= \frac{64\%}{2} = 32\% \text{ each}$$

68% of one lone pair will made by p-orbital.

lone pair = 32% + 68% = 100.

Option (D) is correct.

11. **Ans (B)**

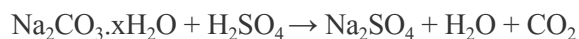


14. **Ans (A)**

$$U_{av} = \frac{\sqrt{\frac{8}{\pi}}}{\sqrt{3}} \times u_{rms} = 0.92 \times 1.85 \times 10^5 \text{ cm/s}$$

$$Z_1 = \frac{u_{av}}{\lambda} = \frac{0.92 \times 1.85 \times 10^5}{1.75 \times 10^{-5}} \text{ sec}^{-1}$$

16. **Ans (D)**



50 ml, 0.1 M

used = 5 - 2.5



2.5 5m.mole moles of

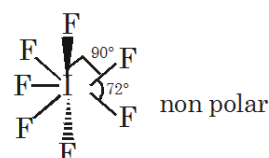
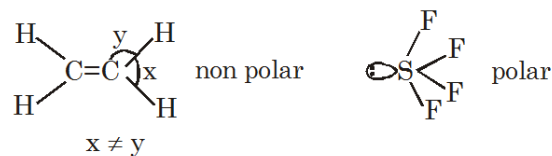
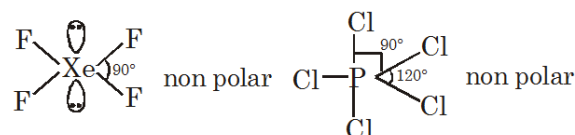
$$Na_2CO_3 \cdot xH_2O = \frac{2.5}{1000} = \frac{0.715}{M}$$

M = 286

PART-2 : CHEMISTRY

SECTION-II

5. **Ans (3.00)**



6. **Ans (180.00)**

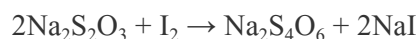
$$P_{1(\text{dry gas})} = P_{\text{wet gas}} - \text{aq. tension} = 740 - 20 = 720 \text{ mm}$$

$$V_1 = 190 \text{ mL}; P_2 = 1 \text{ atm} = 760 \text{ mm}$$

$$P_1 V_1 = P_2 V_2$$

8. **Ans (3.00)**

$$HV = 0.1 \times 12 \text{ mm}$$



$$\text{Millimole of } I_2 = \frac{\eta_{Na_2S_2O_3}}{2} = \frac{1.2}{2}$$

$$\therefore \text{millimole of } Cu^{2+} = 2 \times \frac{1.2}{2}$$

$$\text{Wt. of } Cu^{2+} = 1.2 \times 10^{-3} \times 64 = 0.0768 \text{ gm}$$

$$\therefore \text{by wt. of Cu} = \frac{0.0768}{2.56} \times 100 = 3\%$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)

Image of A(1, 3) in line $x + y = 2$ is

$$\left(1 - \frac{2(2)}{2}, 3 - \frac{2(2)}{2}\right) \equiv (-1, 1)$$

So line BC passes through $(-1, 1)$ and

$$\left(-\frac{2}{5}, -\frac{2}{5}\right)$$

$$\text{Equation of line BC is } y - 1 = \frac{-\frac{2}{5} - 1}{-\frac{2}{5} + 1} (x + 1)$$

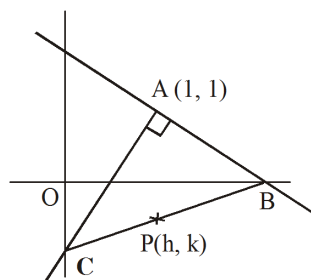
$$\Rightarrow 7x + 3y + 4 = 0$$

2. Ans (A)

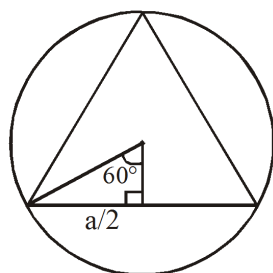
Let $P \equiv (h, k)$, So $B \equiv (2h, 0)$ $C \equiv (0, 2k)$

$$\text{Clearly as } AC \perp AB; \frac{2k-1}{0-1} \times \frac{0-1}{2h-1} = -1$$

$$\Rightarrow 2k - 1 = 1 - 2h. \text{ So, locus is } x + y = 1$$



3. Ans (C)



$$\sin 60^\circ = \frac{p}{a/2}$$

$$a = \frac{2p}{\sin 60^\circ} = \frac{4p}{\sqrt{3}}$$

$$\text{where } p = \frac{1}{\sqrt{2}}$$

$$\frac{p}{R} = \sin 30^\circ$$

$$R = \sqrt{2}$$

$$(x - 1)^2 + (y - 3)^2 = 2$$

4. Ans (A)

Let point be (x_1, y_1)

then polar line $(T = 0)$ $xx_1 + yy_1 = 2$

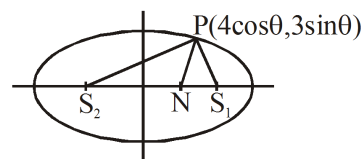
$$\text{comparing with given line } x_1 = y_1 = \frac{2}{5}$$

5. Ans (D)

$(3, x)$ is lying on directrix circle & angle

between tangent is $\frac{\pi}{2}$

6. Ans (A)



$$e = \frac{\sqrt{7}}{4}$$

equation of normal : $4x \sec \theta - 3y \csc \theta = 7$

$$N : \left(\frac{7}{4} \cos \theta, 0\right) \Rightarrow PN = \frac{3}{4} \sqrt{16 - 7 \cos^2 \theta}$$

$$\sqrt{PS_1 \cdot PS_2} = \sqrt{16 - 7 \cos^2 \theta}$$

$$\frac{PN}{\sqrt{PS_1 \cdot PS_2}} = \frac{3}{4}$$

7. Ans (D)

Let $P(t^2, 2t)$ is a point on the parabola then COC is

$$xt^2 + 8ty + 2(x + t^2) + 3 = 0$$

$$\Rightarrow t^2(x + 2) + t(8y) + 2x + 3 = 0$$

$$\Rightarrow \text{curve is } 64y^2 = 4(x + 2)(2x + 3)$$

$$\Rightarrow 16y^2 - 2x^2 - 7x - 6 = 0$$

$$D \neq 0, h^2 > ab \Rightarrow \text{hyperbola}$$

8. Ans (D)

$\therefore$ PQ passes through $(6, 0)$

$$\text{let } P(2t^2, 4t), \text{ then } Q\left(\frac{18}{t^2}, -\frac{12}{t}\right)$$

mid point of PQ $\equiv$ mid point of

$$OR \equiv \left(t^2 + \frac{9}{t^2}, 2t - \frac{6}{t}\right)$$

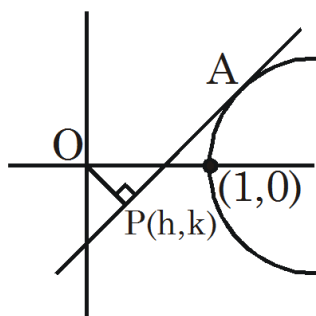
$$\Rightarrow R \equiv \left(2\left(t^2 + \frac{9}{t^2}\right), 4\left(t - \frac{3}{t}\right)\right)$$

$$\therefore \text{locus } y^2 = 16\left(t^2 + \frac{9}{t^2} - 6\right) = 8(x - 12)$$

9. Ans (A)

$$L = 4a \operatorname{cosec}^2 \alpha = 8.5 = 40$$

10. Ans (D)



$$m_{OP} = \frac{k}{h} \Rightarrow m_{PA} = -\frac{h}{k}$$

$$\Rightarrow \text{equation of PA : } y - k = -\frac{h}{k}(x - h)$$

$$\Rightarrow hx + ky = h^2 + k^2 \dots(1)$$

Also PA is tangent to $y^2 = 8(x - 1)$

$\Rightarrow$ equation of

$$PA : y = m(x - 1) + \frac{2}{m}$$

$$\Rightarrow mx - y = m - \frac{2}{m} \dots(2)$$

$$\text{compare (1) and (2) } \frac{m}{h} = \frac{-1}{k} = \frac{m - \frac{2}{m}}{h^2 + k^2}$$

$$\Rightarrow x^3 + xy^2 + 2y^2 - x^2 = 0$$

11. Ans (C)

The equation of the tangent at (4, -2) to

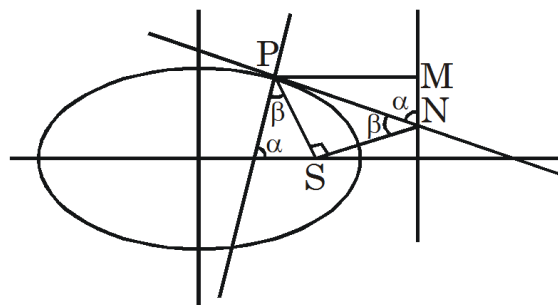
$$y^2 = x \text{ is } -2y = \frac{1}{2}(x + 4) \text{ or } x + 4y + 4 = 0$$

its slope is $-1/4$

Therefore the slope of the perpendicular line is

4. Since the tangents at the end of focal chord of a parabola are at right angle, the slope of the tangent at Q is 4. Hence (C) is the correct answer

12. Ans (C)



$$\sin \alpha = \frac{PM}{PN}$$

$$\sin \beta = \frac{PS}{PN}$$

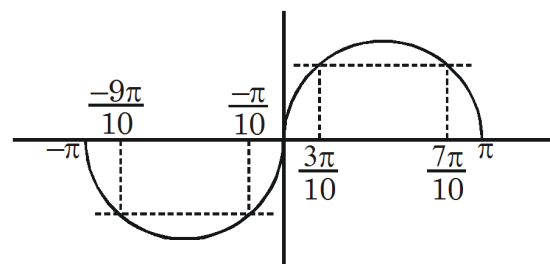
$$\frac{\sin \alpha}{\sin \beta} = \frac{PS}{PM} = \frac{1}{3}$$

$$\sin \beta = \frac{1}{3} \cdot \sin \alpha = \frac{1}{3} \cdot \frac{3}{5} = \frac{1}{5}$$

13. Ans (C)

$$-\sqrt{5} < 4 \sin x - 1 < \sqrt{5}$$

$$\Rightarrow \frac{1 - \sqrt{5}}{4} < \sin x < \frac{\sqrt{5} + 1}{4}$$



$$\therefore -\pi < x < -\frac{9\pi}{10}, \quad -\frac{\pi}{10} < x < \frac{3\pi}{10}, \quad \frac{7\pi}{10} < x < \pi$$

14. Ans (C)

$$\frac{\sin 5^\circ \sin(60^\circ - 5^\circ) \sin(60^\circ + 5^\circ)}{\sin 25^\circ \sin(60^\circ - 25^\circ) \sin(60^\circ + 25^\circ)}$$

$$= \frac{\sin 15^\circ}{\sin 75^\circ} = \frac{\sin 15^\circ}{\cos 15^\circ} = 2 - \sqrt{3}$$

15. Ans (C)

$$ax^2 + bx + c = 0$$

$$x(ax + b) = -c$$

$$ax + b = -\frac{c}{x}$$

$$\frac{1}{(a^2 + \beta^2)} \left[\frac{\alpha^2 + \beta^2}{c^2} \right]$$

16. **Ans (C)**

$$\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ are in A.P.}$$

$$\frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c} \text{ are in A.P.}$$

$$1 + \frac{b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c} \text{ are in A.P.}$$

$$\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c} \text{ are in A.P.}$$

$$\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b} \text{ are in H.P.}$$

17. **Ans (B)**

$$\left| \frac{2k+1}{\sqrt{34}} \right| \leq 1$$

$$\Rightarrow k \in \{-3, -2, -1, 0, 1, 2\}$$

18. **Ans (D)**

$$\Delta = \frac{1}{2}bc \sin A \Rightarrow \sin A = \frac{1}{2} \Rightarrow A = 30^\circ$$

$$\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2} = \sqrt{3}$$

$$B - C = 120^\circ$$

$$B + C = 150^\circ$$

$$\therefore C = \frac{\pi}{12}$$

19. **Ans (D)**

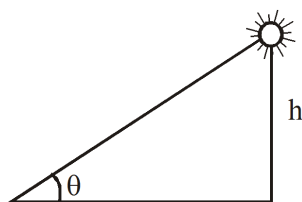
$$\begin{vmatrix} \lambda+5 & \lambda-4 & 1 \\ \lambda-2 & \lambda+3 & 1 \\ \lambda & \lambda & 1 \end{vmatrix} = 0$$

$$\text{Apply } R_1 \rightarrow R_1 - R_2$$

$$R_2 \rightarrow R_2 - R_3$$

$$\Delta = 7 \neq 0$$

20. **Ans (A)**



$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3-2\sqrt{2}}}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{2}-1} = \sqrt{2}+1$$

$$\Rightarrow \theta = \frac{3\pi}{8}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1.00)**

$$\text{RHS} = x$$

$$x^{(\log_2^2 x + 5(\log_2 x) + 5)} = x$$

$$x = 1 \text{ or } (\log_2 x)^2 + 5\log_2 x + 4 = 0$$

$$x_1 = 1, x_2 = \frac{1}{2} \text{ \& } x_3 = \frac{1}{16}$$

$$\Rightarrow x_1 + x_2 + x_3 = \frac{3}{2} + \frac{1}{16} = \frac{25}{16}$$

2. **Ans (2.00)**

$$\therefore 3a = b + c \Rightarrow 4a = a + b + c \Rightarrow s = 2a$$

Now,

$$\cot \frac{B}{2} \cdot \cot \frac{C}{2} = \frac{s(s-b)}{\Delta} \cdot \frac{s(s-c)}{\Delta} = \frac{s(s-b)(s-c)}{s(s-a)(s-b)(s-c)} \cdot s$$

$$= \frac{s}{s-a} = \frac{2a}{2a-a} = 2$$

3. **Ans (5.00)**

Expand determinant

$$\text{so } x^3 + y^3 - 1 + 3xy = 0$$

$$\Rightarrow x^3 + y^3 + (-1)^3 - 3(x)(y)(-1) = 0$$

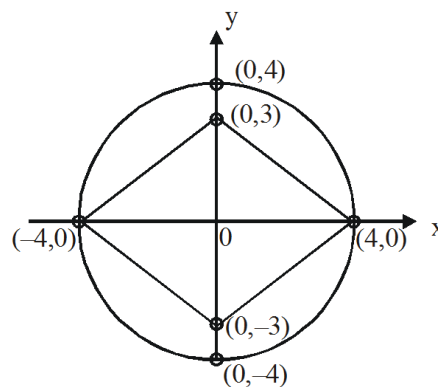
$$\Rightarrow [x + y - 1] [(x - y)^2 + (x + 1)^2 + (y + 1)^2] = 0$$

$$\Rightarrow \text{either } x + y = 1 \text{ or } x = y = -1$$

$$\therefore \text{number of ordered pair } (x, y)$$

$$\equiv (2, -1), (-1, 2), (0, 1), (1, 0), (-1, -1)$$

4. **Ans (2.00)**



5. **Ans (4.00)**

$$\text{Chord of contact} = \left(\frac{\sqrt{3} \cos \theta}{2} \right) x + (\sqrt{3} \sin \theta) y = 1$$

$$\text{Square of distance from origin} = \frac{4}{3} \times \frac{1}{3 \sin^2 \theta + 1}$$

$$\text{Ratio of maximum to minimum} = 4$$

6. **Ans (6.00)**

Equation of normal to the parabola $y^2 = 12x$

with slope -1 is $m = -1$, $a = 3$

$$\Rightarrow y = -x - 2(3)(-1) = 3(-1)^3$$

$$x + y = 9 \quad \text{i.e. } k = 9$$

focus of parabola is $(3, 0)$

$$P = \left| \frac{3-9}{\sqrt{2}} \right| \Rightarrow p^2 = 18$$

7. **Ans (2.00)**

$$A_1 = \frac{a+b}{2}, \quad G_1 = a \cdot \left(\frac{b}{a} \right)^{\frac{1}{3}},$$

$$G_2 = a \cdot \left(\frac{b}{a} \right)^{\frac{2}{3}}$$

$$G_1^3 = a^2b, \quad G_2^3 = b^2a$$

$$G_1G_2 = a^2 \left(\frac{b}{a} \right) = ab$$

$$\Rightarrow \frac{G_1^3 + G_2^3}{G_1G_2A_1} = \frac{ab(a+b).2}{ab.(a+b)} = 2$$

8. **Ans (4.00)**

$$\frac{b^2 \sin 2C + c^2 \sin 2B}{\Delta}$$

$$\Rightarrow \frac{8R^2 \cdot \sin A \sin B \sin C}{\Delta} = \frac{abc}{\Delta \cdot R} = 4$$

9. **Ans (64.00)**

$$D = \begin{vmatrix} \alpha - \beta - \gamma & 2\alpha & 2\alpha \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

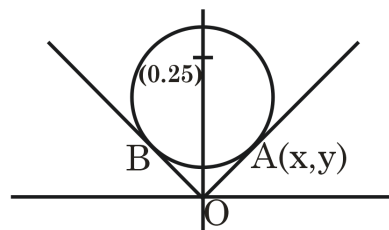
$$D = (\alpha + \beta + \gamma) \begin{vmatrix} 1 & 1 & 1 \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$$

$$C_3 \rightarrow C_3 - C_1 \text{ \& } C_2 \rightarrow C_2 - C_1$$

$$D = (\alpha + \beta + \gamma) \begin{vmatrix} 1 & 0 & 0 \\ 2\beta & -(\alpha + \beta + \gamma) & 0 \\ 2\gamma & 2\gamma & -(\alpha + \beta + \gamma) \end{vmatrix}$$

$$D = (\alpha + \beta + \gamma)^3 = 64$$

10. **Ans (7.00)**



for $\left| \frac{y}{x} \right|$ minimum (x, y) is point of contact of tangent from $(0, 0)$ and $\left| \frac{y}{x} \right|$ is slope of OA or OB

Let equation of OA is $y = mx$

$$\Rightarrow \frac{|25|}{\sqrt{m^2 + 1}} = 15 \Rightarrow (m^2 + 1) = \frac{25}{9}$$

$$m^2 = \frac{16}{9} \quad \left| \frac{y}{x} \right| = |m| = \frac{4}{3}$$